

Bayesian Spatiotemporal Multitask Learning for Radar HRRP Target Recognition

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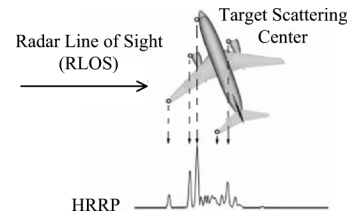


Fig. 1. Illustration of a HRRP sample from a plane target, where the circles on the plane represent some scatterers. This figure is cited from [29].

Abstract—A Bayesian dynamic model based on multitask learning (MTL) is developed for radar automatic target recognition (RATR) using high-resolution range profile (HRRP). The aspect-dependent HRRP sequence is modeled using a truncated stick-breaking hidden Markov model (TSB-HMM) with time-evolving transition probabilities, in which the spatial structure across range cells is described by the hidden Markov structure and the temporal dependence between HRRP samples is described by the time evolution of the transition probabilities. This framework imposes the belief that temporally proximate HRRPs are more likely to be drawn from similar HMMs, while also allowing for possible distant repetition or “innovation”. In addition, as formulated the stick-breaking prior and MTL mechanism are employed to infer the number of hidden states in an HMM and learn the target-dependent states collectively for all targets. The form of the proposed hierarchical model allows efficient variational Bayesian (VB) inference, of interest for large-scale problems. To validate the formulation, example results are presented for an illustrative synthesized dataset and our main application—RATR, for which we consider the measured HRRP data. For the latter, we also make comparisons to the model with the independent state-transition statistics and some other existing statistical models for radar HRRP data.

Index Terms—Hidden Markov model (HMM), hierarchical Bayesian modeling, high-resolution range profile (HRRP), multitask learning (MTL), radar automatic target recognition (RATR), variational Bayes (VB).

I. INTRODUCTION

A high-resolution range profile (HRRP), a real vector, is composed of the amplitude of the coherent summations of the complex returns from target scatterers in each

range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight (LOS), as shown in Fig. 1. Since it contains abundant target information, such as target size, scatterer distribution, etc., radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [4], [6]–[9], [13], [16], [18], [26], [28], [29]. Another approach for RATR is to form a synthetic radar (SAR) or inverse SAR (ISAR) image with subsequent classification performed in the image domain [5], [10], [15], of which the SAR image is usually used for the fixed target and the ISAR image for the moving target, e.g., moving vehicle, airplane or ship. In the following experiments, we mainly discuss the case of flying airplane targets. As discussed in [11], [25], since the moving target is usually noncooperative in ISAR imaging, great difficulties are brought for motion compensation stage when the target’s motion status and trajectory can not be measured precisely (especially for maneuvering airplane target). Therefore, an obvious advantage of HRRP based RATR is that the HRRP data are processed directly without explicitly forming an image in not only training and but also classification phases.

Several literatures [6], [7], [9], [13], [18] showed that statistical recognition is an efficient method for HRRP based RATR. By statistical recognition is meant that the class membership of a measurement HRRP sample $\mathbf{x}$ is determined based on the posterior probabilities of all class memberships $p(c|\mathbf{x})$ ($c \in \{1, \dots, C\}$ denotes class membership). According to Bayes theorem, we have $p(c|\mathbf{x}) \propto p(\mathbf{x}|c)p(c)$, where $p(\mathbf{x}|c)$ is the class-conditional likelihood and $p(c)$ is the prior class probability. Since the prior class probability is usually assumed to be uniformly distributed, estimation of the posterior probability is turned into estimation of the class-conditional likelihood. Some parametric statistical models are developed to describe the class-conditional likelihood of HRRP data, e.g., Gaussian model [13], [18], Gamma model [6], the two-distribution compounded model comprising Gamma distribution and Gaussian mixture distribution [9] and factor analysis (FA) model [7].

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There are usually two stages in the statistical recognition procedure. Firstly, the model parameters are estimated via training data in the training phase; secondly, in the classification phase, as discussed above, given a test sample $\mathbf{x}$, the class-conditional likelihood $p(\mathbf{x}|c)$ can be derived with the estimated parameters for each target c and then $\mathbf{x}$ is associated with target c' if $c' = \arg \max_c p(c|\mathbf{x})$. Nevertheless, these statistical models [6], [7], [9], [13], [18] only characterize the spatial structure across range cells in a single HRRP sample, while ignoring the temporal dependence between HRRP samples.

To make use of the temporal dependence of multiple HRRPs from a sequence of target-sensor orientations, the hidden Markov model (HMM) has been utilized by [16] and [28] for radar HRRP sequence recognition, where each HMM “state” reflects a set of generally contiguous target-sensor orientations over which the HRRP statistics are relatively stationary and the statistical relationships between HRRPs are described via the transition probabilities from one state to the next under HMM formulation. Since it is often difficult to achieve reliable classification based on a single view of the target, multiaspect (sequence) recognition can be a better choice to improve the recognition rates. Nevertheless, there are also some challenges for the experimental system. First, tracking radar is required for the classification phase to get an aspect-dependent sequence at a target, which should be over several states to reflect the aspect-dependent statistics in HMM; second, the angular sampling rates should remain the same for the training and classification phases, since the temporal dependence is used as the discriminative information. In fact, even if the second condition is satisfied, it is still hard to control the detailed temporal dependence between test samples from a moving target, since the target pose is unknown and noncooperative. All these difficulties are brought by utilizing the aspect-dependent HRRP sequence in the classification phase.

The research reported here seeks an alternative approach to characterize both the spatial structure across range cells and the temporal dependence between HRRP samples, in which the multiaspect HRRP sequence is utilized for training the model, but a single HRRP sample for determining its class membership in the classification phase. Specifically, we develop a Bayesian spatiotemporal multitask learning (MTL) model, called *dynamic multitask learning-based HMM with truncated stick-breaking prior* (dynamic MTL-TSB-HMM) for radar HRRP data, where a truncated stick-breaking HMM across range cells is designed for each subset of contiguous HRRP samples from all targets of interest, with “states” shared across all targets’ data and state transition probabilities evolving from subset to subset for a given target. The main characteristics of the proposed model can be summarized as follows:

- *HMM structure across range cells*: Since the “sequential” relationship between the range cells within a single HRRP sample can reflect the physical composition (i.e., scatterers distribution) of the target, we are motivated to use an HMM structure for a single HRRP across range cells. Thus each HMM “state” here represents a kind of scatterers’ echoes and the statistical relationships between echoes from different range cells are described via the transition probabilities from one state to another in a Markov process. Fur-

thermore, since the actual distribution of scatterers on the targets of interest is unknown, the particular set of states that are sampled is deemed “hidden,” motivating an HMM structure. Compared with the previous FA model for HRRP based RATR [7], the HMM structure can avoid the full covariance estimation for the high-dimensional HRRP data, which requires large computation burden, large storage space and large data samples.

- *Non-parametric model selection via stick-breaking construction*: In [16], the number of states is preset and fixed, which can lead to overfitting if set too high or underfitting if set too low. A Bayesian nonparametric, (countably) infinite-state HMM that utilizes the hierarchical Dirichlet process (HDP) [24], has proven effective to infer the optimal number of states in many applications [19], [24]. However, the lack of conjugacy between the two levels of Dirichlet processes means that a truly variational solution [1] is difficult, making this approach computationally prohibitive when modeling very large sequential datasets. Recent paper [20] proposes a fully conjugate prior for an infinite-state HMM, where each row of the transition matrix is given an infinite dimensional stick-breaking prior [12], referred to as the stick-breaking HMM (SB-HMM). The truncated version of such stick-breaking construction is utilized in our model for characterizing the HMM states, which is referred to as the truncated stick-breaking HMM (TSB-HMM).
- *MTL*: A limitation of statistical model is that it usually requires substantial training data, assumed to be similar to the data on which the model is tested. However, in RATR problems, one may have limited training data. In addition, the test data may be obtained under different motion circumstance. Rather than building models for each data subset associated with different target-aspect individually, it is desirable to appropriately share the information among these related data, thus offering the potential to improve overall recognition performance. If the modeling of one data subset is termed one learning task, the learning of models for all tasks jointly is referred to as multitask learning (MTL) [3]. We here extend the TSB-HMM learning in a multitask setting for radar HRRP recognition. Moreover, such MTL mechanism makes it possible to further exploit the statistical interrelationships between HRRP samples, i.e., the temporal dependence.
- *Time-evolution structure across HRRP samples*: The state transition probabilities of HMMs are time-evolving across the HRRP sequence in our framework, which explicitly imposes the belief that HRRP samples that are temporally (orientationally) proximate are more likely to be drawn from HMMs with similar parameters, while also allowing for possible distant repetition (i.e., data being similar in separated angles which might be led by symmetries in the objects [6]) or “innovation” (i.e., new HMMs occur at some aspects which might be associated with the abrupt fluctuation in the HRRP sequence). Recently, [17], [22], [23] propose a Bayesian time-evolution structure with application to document analysis, gene analysis and music analysis, respectively. Based on these work, a similar dynamic

structure is considered here to incorporate the temporal dependence between HRRP samples into the MTL-HMM structure. Our model differs from the dynamic HDP-HMM (dHDP-HMM) for music analysis [23] in that our HMM states do not evolve over tasks; it is the transition matrices over states that temporally change, while the overall set of states keeps unchanged. Recalling that each HMM “state” in our model represents a kind of scatterers’ echoes, specific state-transition statistics are typically localized over an aspect sector for a given target, which is a reasonable assumption for radar HRRP data.

- *Full VB inference:* We present a fully conjugate Bayesian model structure, which does have an efficient variational Bayesian (VB) solution.

The remainder of the paper is organized as follows. A brief review of HMM is provided in Section II. We introduce the proposed model for HRRP data in Section III. The model learning and classification methods are discussed in Section IV. The detailed experimental results based on the synthetic data and measured HRRP data are provided in Section V, followed by conclusions in Section VI.

II. REVIEW OF HMM

The HMM is a generative statistical representation of sequential data [21]. Given an observation sequence $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T)$ with $\mathbf{x}_t \sim f(\boldsymbol{\theta}_{s_t})$, where $f(\cdot)$ represents the observation model, the corresponding hidden sequence is $\mathbf{s} = (s_1, s_2, \dots, s_T)$, from which $s_t \in \{1, \dots, I\}$ with I denoting the number of states and $\boldsymbol{\theta}_{s_t}$ is the set of parameters for the observation model indexed by the state at time t , s_t . An HMM is represented by parameters $\Omega = \{\mathbf{w}_0, \mathbf{W}, \boldsymbol{\theta}\}$, defined as

- $\mathbf{w}_0 = \{w_{0,i}\}$, $w_{0,i} = Pr(s_1 = i)$: initial state distribution;
- $\mathbf{W} = \{w_{i,j}\}$, $w_{i,j} = Pr(s_{t+1} = j | s_t = i)$: state transition probability;
- $\boldsymbol{\theta} = \{\boldsymbol{\theta}_i\}$, $\mathbf{x}_t \sim f(\boldsymbol{\theta}_{s_t})$: emission likelihood.

For a given Ω , the joint probability of the observation and the underlying state sequence is expressed as

$$p(\mathbf{x}, \mathbf{s} | \mathbf{W}, \boldsymbol{\theta}, \mathbf{w}_0) = w_{0,s_1} \prod_{t=1}^{T-1} w_{s_t, s_{t+1}} \prod_{t=1}^T p(\mathbf{x}_t | \boldsymbol{\theta}_{s_t}). \quad (1)$$

The data likelihood $p(\mathbf{x} | \mathbf{W}, \boldsymbol{\theta}, \mathbf{w}_0)$ can be obtained by integrating over the states using the forward algorithm [21]. The priors associated with $\mathbf{w}_0$ and the rows of $\mathbf{W}$ are typically Dirichlet distributions, since these are conjugate to the multinomial likelihood. In [21], the number of states associated with an HMM is initialized and fixed.

The stick-breaking prior [12] can be used to develop an HMM with an unknown number of states [20]. For this stick-breaking HMM (SB-HMM), each row of the infinite state transition matrix, $\mathbf{W}$, is given a stick-breaking prior.

$$\begin{aligned} G_i &= \sum_{j=1}^{\infty} w_{i,j} \delta(\boldsymbol{\theta}_j), \quad w_{i,j} = v_{i,j} \prod_{h=1}^{j-1} (1 - v_{i,h}), \\ v_{i,j} | \alpha_i &\sim \text{Beta}(1, \alpha_i), \quad \boldsymbol{\theta}_j \sim H, \\ \alpha_i | a_\alpha, b_\alpha &\sim \text{Ga}(a_\alpha, b_\alpha); \quad i = 1, 2, \dots, \infty; \\ j &= 1, 2, \dots, \infty \end{aligned} \quad (2)$$

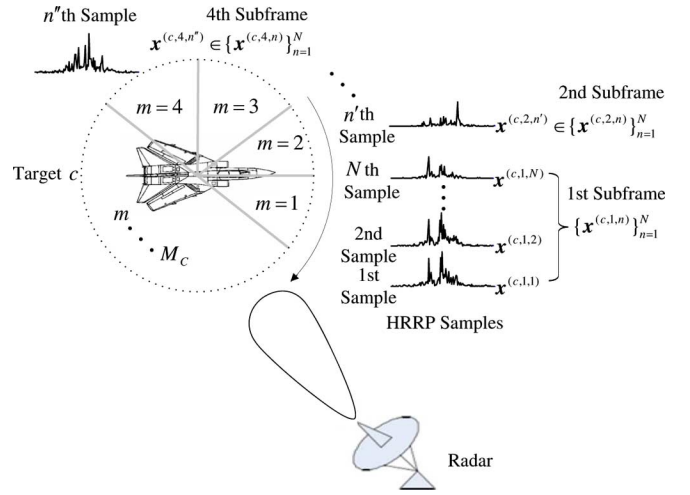


Fig. 2. Illustration of the subframe partition for a set of HRRP samples from the plane target c , where the aircraft is assumed to rotate around radar with a given depression angle and the samples within a small aspect-angular sector is referred to as a subframe. Here m is the subframe index, M_c denotes the number of subframes for target c , n is the sample index in a subframe, N denotes the number of samples in a subframe and $\mathbf{x}^{(c,m,n)}$ denotes the n th HRRP sample in the m th subframe of target c .

where G_i denotes an infinite mixture distribution for state i with a set of state-dependent parameters $\{\boldsymbol{\theta}_j\}_{j=1}^{\infty}$ and respective mixture weights $\{w_{i,j}\}_{j=1}^{\infty}$, $\delta(\boldsymbol{\theta}_j)$ is a point measure concentrated at $\boldsymbol{\theta}_j$, $\text{Beta}(1, \alpha_i)$ represents the Beta distribution with hidden variable α_i , the drawn variables $\{v_{i,j}\}_{j=1}^{\infty}$ are independent and identically distributed (i.i.d.), $\text{Ga}(a_\alpha, b_\alpha)$ represents the Gamma distribution with parameters a_α and b_α and H denotes a prior distribution from which the set $\{\boldsymbol{\theta}_j\}_{j=1}^{\infty}$ is i.i.d. drawn. The initial state probability mass function (pmf), $\mathbf{w}_0$, is also constructed according to an infinite stick-breaking construction. The key advantage in this formulation is that the state transition pmf, $\mathbf{w}_i$, for state i has an infinite stick-breaking prior on the domain of the positive integers, which index the states; while the corresponding state-dependent parameters, $\boldsymbol{\theta}_j$, are drawn separately, effectively detaching the construction $\boldsymbol{\theta}$ from the construction $\mathbf{W}$. This is in contrast with Dirichlet distribution priors, where these two matrices are linked in their construction, thus requiring the two-level hierarchical Dirichlet process (HDP) solution for the HDP-HMM structure [24].

The formulation in (2) is infinite, however, when the i th row of the state transition matrix $\mathbf{w}_i$, terminates at some finite number $I - 1$ with $w_{i,I} \equiv 1 - \sum_{j=1}^{I-1} w_{i,j}$, the result is a draw from a generalized Dirichlet distribution (GDD) [27]. To simplify notation below, constructing the truncated form of $\mathbf{w}_i$ is denoted as $\mathbf{w}_i \sim \text{GDD}(\mathbf{1}_{(I-1) \times 1}, [\alpha_i]_{(I-1) \times 1})$, where $\mathbf{1}_{(I-1) \times 1}$ denotes an $(I - 1)$ -length vector of ones, $[\alpha_i]_{(I-1) \times 1}$ denotes an $(I - 1)$ -length vector of α_i and I is the truncation number of states (usually a large number). Such truncated stick-breaking priors over the state transitions in an HMM are necessary for the variational inference and the corresponding truncated version of HMM is referred to as the truncated stick-breaking HMM (TSB-HMM). Although there are fixed I states in the TSB-HMM, only a subset of these states will contain substantial weights in the posterior as discussed in [20].

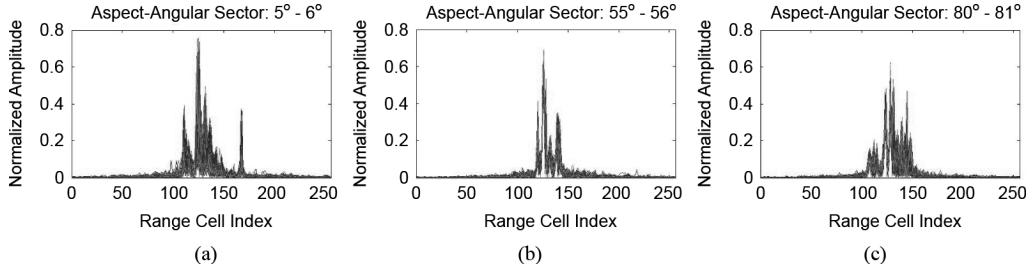


Fig. 3. HRRP samples of An-26 in different aspect-angular sectors (a) aspect-angular sectors: 5° – 6° ; (b) aspect-angular sectors: 55° – 56° ; (c) aspect-angular sectors: 80° – 81° . Here the different shades of gray code the different aspect angles.

III. BAYESIAN SPATIOTEMPORAL MTL MODEL

A. Model Construction

For the motivating RATR problem of interest here, one typically has access to multispect HRRP datasets from multiple targets. As shown in Fig. 2, to model a multispect HRRP dataset from target c ($c \in \{1, \dots, C\}$ with C denoting the number of targets) here, we divide the HRRP dataset into M_c subsets, e.g., the m th dataset (here $m \in \{1, \dots, M_c\}$) is $\{\mathbf{x}^{(c,m,n)}\}_{n=1}^N$ with N denoting the number of samples, $\mathbf{x}^{(c,m,n)} = [x^{(c,m,n)}(1), \dots, x^{(c,m,n)}(L)]^T$ representing the n th HRRP sample in this dataset and L denoting the number of range cells in an HRRP sample. According to the scattering center model [11], HRRPs from complex targets yield target signatures that are a strong function of the target-sensor orientation [6]–[9], [13], [16], [18], [28], referred to as target-aspect sensitivity in [7]–[9]. Nevertheless, as shown in Fig. 3, the HRRPs within a small aspect-angular sector change little and can be assumed to be i.i.d., therefore, the spatial structure is assumed to be same for all $\{\mathbf{x}^{(c,m,n)}\}_{n=1}^N$ in a given subset with $m \in \{1, \dots, M_c\}$ and $c \in \{1, \dots, C\}$. We refer to such subset as a *subframe* in this paper. Here we wish to learn a TSB-HMM for each of these subframes to characterize the spatial structure across range cells and further exploit the temporal dependence between these subframes. If the learning of TSB-HMM for the m th subframe from target c , i.e., $\{\mathbf{x}^{(c,m,n)}\}_{n=1}^N$, is termed the learning task m from target c , the learning of TSB-HMMs for all C targets, each with M_c subframes, jointly is referred to as MTL [3].

In the MTL-TSB-HMM mechanism for HRRP data, since each HMM “state” here represents a kind of scatterers’ echoes and the state-transition statistics impact the probability of manifesting particular states, we here assume these transition probabilities are aspect-dependent, but the observation statistics do *not* depend on which specific target or subframe is being considered. So motivated, in the proposed model, the state-transition statistics are assumed to be time-evolving, but the state-dependent observation statistics are shared across all $\sum_{c=1}^C M_c$ subframes. To constitute an HMM structure with time-evolving state-transition statistics across all subframes from a given target, the state-transition statistics of the m th subframe for target c , defined by $\mathbf{W}^{(c,m)}$, is modified from $\mathbf{W}^{(c,m-1)}$ by introducing a new innovation state-transition matrix $\mathbf{A}^{(c,m)}$ and a random variable $\rho_m^{(c)} \in [0, 1]$ controls the degree of innovation in $\mathbf{W}^{(c,m)}$ relative to $\mathbf{W}^{(c,m-1)}$. Specifically,

the proposed model, termed the dynamic MTL-TSB-HMM, is represented as

$$\begin{aligned}
 x^{(c,m,n)}(l) &\sim f(\theta_{s_l^{(c,m,n)}}); l = 1, \dots, L; \\
 n &= 1, \dots, N; m = 1, \dots, M_c; c = 1, \dots, C \\
 s_l^{(c,m,n)} &\sim \begin{cases} \mathbf{w}_0^{(c,m)} & \text{if } l = 1 \\ \mathbf{w}_{s_{l-1}^{(c,m,n)}}^{(c,m)} & \text{if } l > 1 \end{cases} \\
 \mathbf{W}^{(c,m)} &= [\mathbf{w}_0^{(c,m)^T}; \mathbf{w}_1^{(c,m)^T}; \dots; \mathbf{w}_I^{(c,m)^T}] \\
 \mathbf{A}^{(c,m)} &= [\mathbf{a}_0^{(c,m)^T}; \mathbf{a}_1^{(c,m)^T}; \dots; \mathbf{a}_I^{(c,m)^T}] \\
 \begin{cases} \mathbf{W}^{(c,m)} = \mathbf{A}^{(c,m)} \\ \mathbf{W}^{(c,m)} = (1 - \rho_m^{(c)})\mathbf{W}^{(c,m-1)} + \rho_m^{(c)}\mathbf{A}^{(c,m)} \\ \rho_m^{(c)} \sim \text{Beta}(1, f_0) \end{cases} & \begin{aligned} &\text{if } m = 1 \\ &\text{if } m > 1 \end{aligned} \\
 \mathbf{a}_i^{(c,m)} | \alpha_i &\sim \text{GDD}(\mathbf{1}_{(I-1) \times 1}, [\alpha_i]_{(I-1) \times 1}); i = 0, \dots, I \\
 \alpha_i | a_\alpha, b_\alpha &\sim \text{Ga}(a_\alpha, b_\alpha); i = 0, \dots, I \\
 \theta_i &\sim H; i = 1, \dots, I
 \end{aligned} \tag{3}$$

where $x^{(c,m,n)}(l)$ denotes the n th sample’s amplitude echo in the l th range cell in the m th subframe of the c th target; $s_l^{(c,m,n)}$ is the corresponding state indicator; a_α , b_α and f_0 are preset hyper-parameters. The graphical representation Fig. 4(a) indicates that we learn TSB-HMMs with aspect-dependent state transition matrices for all of the subframes from a given target as well as share the same state set across the subframes from all targets; while Fig. 4(b) shows that the spatial dependence across range cells for a given subframe is characterized via an HMM structure. Therefore, a *state* here reflects the amplitude of a range cell rather than the aspect-angle of an HRRP in [16]. According to the scattering center model, paper [9] roughly categorizes range cells into three kinds: i) a large number of small scatterers and no predominant scatterers; ii) a predominant scatterer and a large number of small scatterers; iii) a large number of small scatterers and several predominant scatterers. By contrast with [9], here HMM states can more exactly characterize different kinds of scatterers’ echoes, especially for the third type of range cell in [9]. In addition, the HMM structure here can further describe the spatial relationship between range cells, which can reflect the physical composition (i.e., scatterers distribution) of a target.

The form of the observation model $f(\cdot)$ may vary depending on the application; in the work presented here, since the Normal distribution are used to describe HRRP data in [7], [13], [18] with good performance, $f(\cdot)$ is defined as a univariate Normal

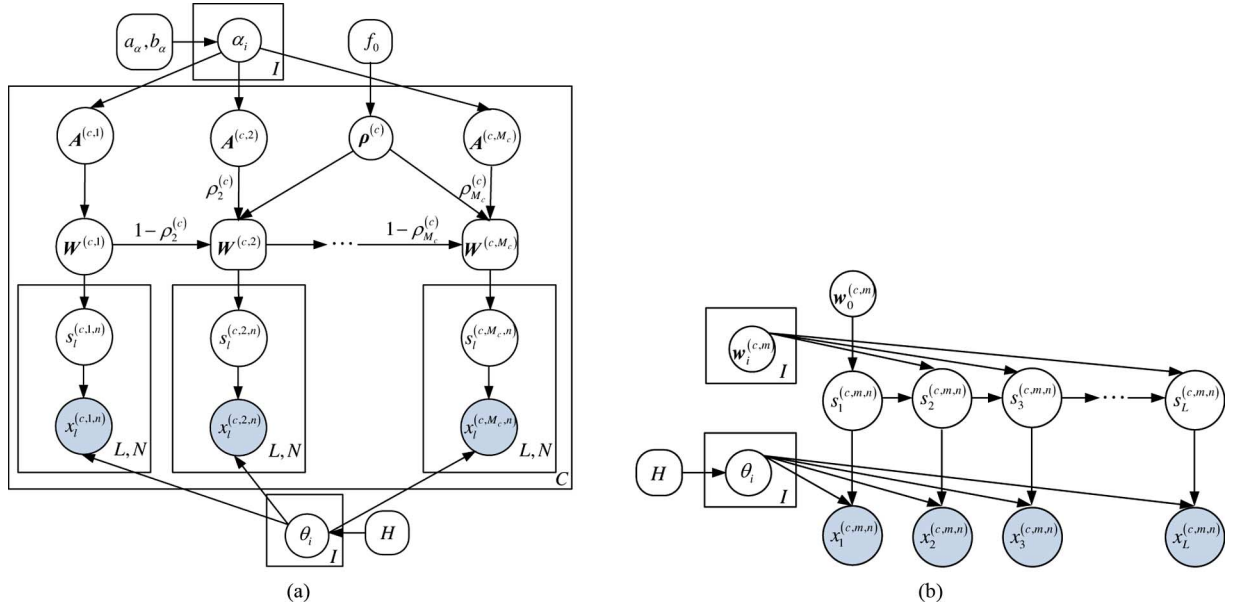


Fig. 4. Graph of the dynamic MTL-TSB-HMM. The detailed generative process is described in (3). (a) Graphical representation of the general model, where the vector $\rho^{(c)} = [\rho_1^{(c)}, \dots, \rho_{M_c}^{(c)}]^T$; (b) the expanded representation of the HMM structure for the m th subframe from target c , where $[w_0^{(c,m)T}; w_1^{(c,m)T}; \dots; w_L^{(c,m)T}] = W^{(c,m)}$. Here i is the state index, c is the target index, l is the range cell index, n is the sample index in a subframe, m is the subframe index and $x_l^{(c,m,n)}$ denotes the target echo in the l th range cell from n th HRRP sample in the m th subframe of target c . For the symbol L, N, C or I at the edge of a box indicates that there is a stack of such boxes for the index l, n, c , or i , where L denotes the number of range cells in an HRRP sample, N denotes the number of samples in a subframe, C denotes the number of targets, I denotes the truncation number of states.

distribution, and the corresponding $H(\cdot)$ is a Normal-Gamma distribution to preserve the conjugacy requirements. Nevertheless, it is important to point out that such univariate Normal distribution assumption is different from the independent Gaussian model used in [13] and [18], since the HMM structure actually describes the sequential dependence across elements in an HRRP sample.

B. Dynamic Structure Analysis

The dynamic structure in (3) (actually the temporal dependence) can further be extended

$$\begin{aligned}
 W^{(c,m)} &= (1 - \rho_m^{(c)})W^{(c,m-1)} + \rho_m^{(c)}A^{(c,m)} \\
 &= \prod_{k=2}^m (1 - \rho_k^{(c)})A^{(c,1)} \\
 &\quad + \sum_{k=2}^m \left\{ \prod_{h=k+1}^m (1 - \rho_h^{(c)}) \right\} \rho_k^{(c)} A^{(c,k)} \\
 &= u_{m,1}^{(c)} A^{(c,1)} + u_{m,2}^{(c)} A^{(c,2)} \\
 &\quad + \dots + u_{m,m}^{(c)} A^{(c,m)} \quad m = 2, \dots, M_c \quad (4)
 \end{aligned}$$

where $u_{m,k}^{(c)} = \rho_k^{(c)} \prod_{h=k+1}^m (1 - \rho_h^{(c)})$, for $k = 1, \dots, m$, with $\rho_1^{(c)} = 1$. It can easily verify that $\sum_{k=1}^m u_{m,k}^{(c)} = 1$ for each m , with $u_{m,k}^{(c)}$ denoting the prior probability that state-transition matrix $W^{(c,m)}$ is drawn from the $\{A^{(c,1)}, A^{(c,2)}, \dots, A^{(c,m)}\}$.

The global HMM and independent MTL-HMM across all subframes from each target are the limiting cases of this model:

- when $\rho_m^{(c)} \rightarrow 0$, $W^{(c,m)} \rightarrow W^{(c,m-1)}$ and there is no innovation, resulting in a common state-transition matrix for all subframes from target c ;
- when $\rho_m^{(c)} \rightarrow 1$, $W^{(c,m)} \rightarrow A^{(c,m)}$, where the new innovation $A^{(c,m)}$ controls the time-evolution, resulting in each subframe from target c having a unique state-transition matrix.

Here we should restate again that the dynamic MTL-TSB-HMM does not assume the HMM “states” evolve over time, only the transition matrix. The states are shared explicitly across all subframes. This is in contrast with the dHDP-HMM in [23], where the similar dynamic structure is imposed on HDP-HMM, thus resulting each mixture component is an independent HMM and the mixing weights of these components evolve over time.

IV. MODEL LEARNING AND CLASSIFICATION

A. Alternative Representation

In Bayesian inference, (4) is usually realized via an extra indicator variable $z_m^{(c)}$ drawn from the discrete pmf $\mathbf{u}_m^{(c)} = [u_{m,1}^{(c)}, \dots, u_{m,m}^{(c)}]^T$ to indicate which state-transition matrix is taken from $\{A^{(c,1)}, A^{(c,2)}, \dots, A^{(c,m)}\}$ as $W^{(c,m)}$ for subframe m . Since different subframes from a given target might select a same state-transition matrix, these subframes can combine a larger set that share the exactly same HMM. In this paper, such combined larger set is referred to as *frame* to distinguish from *subframe* and the variable $z_m^{(c)}$ is referred to as

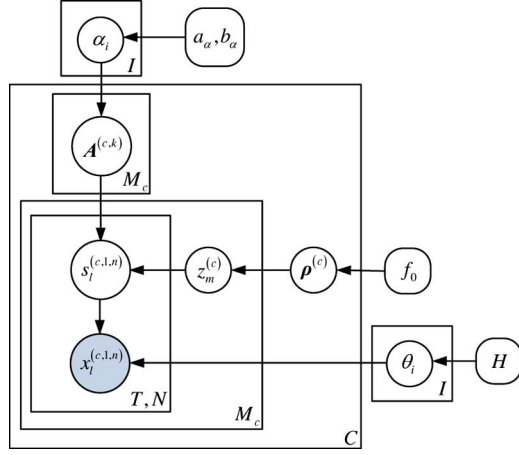


Fig. 5. Graphical representation of the dynamic MTL-TSB-HMM from a stick-breaking point of view. This is an alternative form of the model in Fig. 4, of which the detailed generative process is described in (5). Here $A^{(c,k)} = [a_0^{(c,k)^T}; a_1^{(c,k)^T}; \dots; a_I^{(c,k)^T}]$.

the frame indicator for subframe m of target c . An alternative form of the dynamic MTL-TSB-HMM model is represented as

$$\begin{aligned}
 x^{(c,m,n)}(l) &\sim f(\theta_{s_l^{(c,m,n)}}); l = 1, \dots, L; \\
 n &= 1, \dots, N; m = 1, \dots, M_c; c = 1, \dots, C \\
 s_l^{(c,m,n)} &\sim \begin{cases} a_0^{(c,z_m^{(c)})} & \text{if } l = 1 \\ a_{s_{l-1}^{(c,m,n)}}^{(c,z_m^{(c)})} & \text{if } l > 1 \end{cases} \\
 z_m^{(c)} &\sim \mathbf{u}_m^{(c)}, u_{m,k}^{(c)} = \rho_k^{(c)} \prod_{h=k+1}^m (1 - \rho_h^{(c)}); k = 1, \dots, m \\
 \rho_1^{(c)} &= 1; \rho_m^{(c)} \sim \text{Beta}(1, f_0); m = 2, \dots, M_c \\
 \mathbf{a}_i^{(c,k)} | \alpha_i &\sim \text{GDD}(\mathbf{1}_{(I-1) \times 1}, [\alpha_i]_{(I-1) \times 1}); i = 0, \dots, I \\
 \alpha_i | a_\alpha, b_\alpha &\sim \text{Ga}(a_\alpha, b_\alpha); i = 0, \dots, I \\
 \theta_i &\sim H; i = 1, \dots, I
 \end{aligned} \tag{5}$$

and the graphical representation is shown in Fig. 5. In this framework, we first draw the frame indicator variable $z_m^{(c)}$ from a stick-breaking construction and then the state indicator $\{s_l^{(c,m,n)}\}_{l=1, n=1}^{L, N}$ from the corresponding $A^{(c,z_m^{(c)})} = [a_0^{(c,z_m^{(c)})^T}; a_1^{(c,z_m^{(c)})^T}; \dots; a_I^{(c,z_m^{(c)})^T}]$ according to the HMM structure. In this way, a TSB-HMM is designed for each subframe of all targets with state shared across all targets' data and state transition probabilities evolving from subframe to subframe for a given target.

B. Posterior Computation

In the training phase, we seek to estimate the posterior distribution of the latent variables Ψ in the model, given the observed training data $\mathbf{X}$ and hyperparameters $\mathbf{T}$, via Bayesian inference

$$p(\Psi | \mathbf{X}, \mathbf{T}) = \frac{p(\mathbf{X} | \Psi, \mathbf{T}) p(\Psi | \mathbf{T})}{\int p(\mathbf{X} | \Psi, \mathbf{T}) p(\Psi | \mathbf{T}) d\Psi} \tag{6}$$

where the denominator $\int p(\mathbf{X} | \Psi, \mathbf{T}) p(\Psi | \mathbf{T}) d\Psi = p(\mathbf{X} | \mathbf{T})$ is the model evidence (marginal likelihood). Variational Bayesian (VB) inference seeks a variational distribution $q(\Psi)$ to ap-

proximate the true posterior distribution of the latent variables $p(\Psi | \mathbf{X}, \mathbf{T})$. The expression

$$\begin{aligned}
 \log p(\mathbf{X} | \mathbf{T}) &= \mathcal{L}(q(\Psi)) + \mathcal{KL}(q(\Psi) \| p(\Psi | \mathbf{X}, \mathbf{T})) \\
 &= \int q(\Psi) \log \frac{p(\mathbf{X} | \Psi, \mathbf{T}) p(\Psi | \mathbf{T})}{q(\Psi)} d\Psi \\
 &\quad + \int q(\Psi) \log \frac{q(\Psi)}{p(\Psi | \mathbf{X}, \mathbf{T})} d\Psi
 \end{aligned} \tag{7}$$

where $\mathcal{L}(q(\Psi)) = \int q(\Psi) \log \frac{p(\mathbf{X} | \Psi, \mathbf{T}) p(\Psi | \mathbf{T})}{q(\Psi)} d\Psi$ forms a lower bound for $\log p(\mathbf{X} | \mathbf{T})$. From (7) $\log p(\mathbf{X} | \mathbf{T}) \geq \mathcal{L}(q(\Psi))$, since the Kullback-Leibler (KL) divergence $\mathcal{KL}(q(\Psi) \| p(\Psi | \mathbf{X}, \mathbf{T})) = \int q(\Psi) \log \frac{q(\Psi)}{p(\Psi | \mathbf{X}, \mathbf{T})} d\Psi$ is nonnegative, which measures the similarity between the two distributions. Accordingly, the goal of minimizing $\mathcal{KL}(q(\Psi) \| p(\Psi | \mathbf{X}, \mathbf{T}))$, i.e., maximizing the similarity between the variational distribution and the true posterior, reduces to adjusting $q(\Psi)$ to maximize $\mathcal{L}(q(\Psi))$.

VB inference [1] assumes a factorized $q(\Psi)$, i.e., $q(\Psi) = \prod_k q_k(\Psi_k)$, typically with the same form as employed in $p(\Psi | \mathbf{X}, \mathbf{T})$. With such an assumption, the variational distributions can be updated iteratively to increase the lower bound $\mathcal{L}(q(\Psi))$. The mean-field variational distribution for (5) is

$$\begin{aligned}
 q(\Psi) &= \left[\prod_{c=1}^C \prod_{k=1}^{M_c} q(A^{(c,k)}) \right] q(\alpha) \\
 &\quad \times \left[\prod_{c=1}^C \prod_{m=1}^{M_c} \prod_{n=1}^N \prod_{l=1}^L q(s_l^{(c,m,n)}) \right] \\
 &\quad \times \left[\prod_{c=1}^C \prod_{m=2}^{M_c} q(z_m^{(c)}) \right] \left[\prod_{c=1}^C q(\rho^{(c)}) \right] [q(\theta)]
 \end{aligned} \tag{8}$$

where $\{\{A^{(c,k)}\}_{c=1, k=1}^{C, M_c}, \alpha, \{s_l^{(c,m,n)}\}_{c=1, m=1, n=1, l=1}^{C, M_c, N, L}, \{z_m^{(c)}\}_{c=1, m=2}^{C, M_c}, \{\rho^{(c)}\}_{c=1}^C, \theta\}$ are the latent variables in this model. In addition, the hyperparameters are $\{a_\alpha, b_\alpha\}$ for α , f_0 for $\{\rho^{(c)}\}_{c=1}^C$, and those for θ , which depend on the specific prior H .

A general method for performing variational inference for conjugate-exponential Bayesian networks is as follows: For a given node in a graphic model, write out the posterior as though everything were known, take the logarithm, the expectation with respect to all unknown parameters and exponentiate the result. The algorithm iteratively updates $q(\Psi)$ so that $\mathcal{L}(q(\Psi))$ approaches to $\log p(\mathbf{X} | \mathbf{T})$ until convergence. The derivation of the update equations for the variational posteriors used in implementing our model is summarized as follows:

1) The updating of $q(\theta_i)$ has the form

$$\begin{aligned}
 \log q(\theta_i) &\propto \log H(\theta_i | \tilde{\theta}_i) \\
 &\quad + \sum_{c=1}^C \sum_{m=1}^{M_c} \sum_{n=1}^N \sum_{l=1}^L \langle s_{l,i}^{(c,m,n)} \rangle \log f(x_l^{(c,m,n)} | \theta_i); \\
 i &= 1, \dots, I
 \end{aligned} \tag{9}$$

where $\tilde{\theta}_i$ denotes the parameters in the posterior for $H(\cdot)$ and $\langle s_{l,i}^{(c,m,n)} \rangle$ denotes the expected number of state indi-

cator $s_i^{(c,m,n)}$ with outcome i . If the prior $H(\cdot)$ is conjugate to the likelihood $f(\cdot)$ then we can easily get the update equation for $q(\theta_i)$. For our application, the specific definition and derivation for $q(\theta_i)$ can be referred to [14].

- 2) For updating $q(\mathbf{a}_0^{(c,k)})$ and $q(\mathbf{a}_i^{(c,k)})$ with $k = 1, \dots, M_c$, $c = 1, \dots, C$ and $i = 1, \dots, I$ given the prior $p(\mathbf{a}_0^{(c,k)}) = \text{GDD}(\mathbf{1}_{(I-1) \times 1}, [\alpha_0]_{(I-1) \times 1})$ and $p(\mathbf{a}_i^{(c,k)}) = \text{GDD}(\mathbf{1}_{(I-1) \times 1}, [\alpha_i]_{(I-1) \times 1})$, assume $q(\mathbf{a}_0^{(c,k)}) = \text{GDD}(\tilde{\alpha}_0^{(c,k)}|_1, \tilde{\alpha}_0^{(c,k)}|_2)$ and $q(\mathbf{a}_i^{(c,k)}) = \text{GDD}(\tilde{\alpha}_i^{(c,k)}|_1, \tilde{\alpha}_i^{(c,k)}|_2)$. Then

$$\begin{cases} \tilde{\alpha}_{0,j}^{(c,k)}|_1 = 1 + \sum_{c=1}^C \sum_{m=1}^{M_c} \langle z_{m,k}^{(c)} \rangle \sum_{n=1}^N \langle s_{1,j}^{(c,m,n)} \rangle \\ \tilde{\alpha}_{0,j}^{(c,k)}|_2 = \frac{\tilde{a}_{\alpha}^{(0)}}{\tilde{b}_{\alpha}^{(0)}} + \sum_{c=1}^C \sum_{m=1}^{M_c} \langle z_{m,k}^{(c)} \rangle \\ \quad \sum_{n=1}^N \sum_{h=j+1}^I \langle s_{1,h}^{(c,m,n)} \rangle \\ j = 1, \dots, I-1 \\ \tilde{\alpha}_{i,j}^{(c,k)}|_1 = \frac{1 + \sum_{c=1}^C \sum_{m=1}^{M_c} \langle z_{m,k}^{(c)} \rangle}{\sum_{n=1}^N \sum_{l=2}^I \langle s_{l-1,i}^{(c,m,n)} \rangle \langle s_{l,j}^{(c,m,n)} \rangle} \\ \tilde{\alpha}_{i,j}^{(c,k)}|_2 = \frac{\tilde{a}_{\alpha}^{(i)}}{\tilde{b}_{\alpha}^{(i)}} + \sum_{c=1}^C \sum_{m=1}^{M_c} \langle z_{m,k}^{(c)} \rangle \\ \quad \sum_{n=1}^N \sum_{l=2}^I \sum_{h=j+1}^I \langle s_{l-1,i}^{(c,m,n)} \rangle \langle s_{l,h}^{(c,m,n)} \rangle \\ i = 1, \dots, I; j = 1, \dots, I-1 \end{cases} \quad (10)$$

where $\langle z_{m,k}^{(c)} \rangle$ denotes the expected number of binary switch indicator $z_m^{(c)}$ with outcome k .

- 3) Assume $q(\alpha_i) = \text{Ga}(\tilde{a}_{\alpha}^{(i)}, \tilde{b}_{\alpha}^{(i)})$, the update equations for $q(\alpha_i)$ are as follows:

$$\begin{cases} \hat{a}_{\alpha}^{(i)} = a_{\alpha} + (\sum_{c=1}^C M_c)(I-1) \\ \hat{b}_{\alpha}^{(i)} = b_{\alpha} - \sum_{c=1}^C \sum_{k=1}^{M_c} \sum_{j=1}^{I-1} \left[\frac{\psi(\tilde{\alpha}_{i,j}^{(c,k)}|_2)}{-\psi(\tilde{\alpha}_{i,j}^{(c,k)}|_1 + \tilde{\alpha}_{i,j}^{(c,k)}|_2)} \right] \\ i = 1, \dots, J \end{cases} \quad (11)$$

in which $\psi(\cdot)$ is the Digamma function.

- 4) Given the approximate distribution of the other variables

$$\begin{aligned} q(\mathbf{s}^{(c,m,n)}) &\propto \exp \left[\sum_{k=1}^{M_c} \langle z_{m,k}^{(c)} \rangle \left[\langle \log a_{0,s_1}^{(c,k)} \rangle \right. \right. \\ &\quad + \sum_{l=1}^{L-1} \langle \log a_{s_l^{(c,m,n)}, s_{l+1}^{(c,m,n)}}^{(c,k)} \rangle \\ &\quad \left. \left. + \sum_{l=1}^L \langle \log f(x_l^{(c,m,n)} | \theta_{s_l^{(c,m,n)}}) \rangle \right] \right] \\ n = 1, \dots, N; m = 1, \dots, M_c; c = 1, \dots, C \end{aligned} \quad (12)$$

where $\langle \cdot \rangle$ represents the expectation of the associated variable's distribution. One may readily derive that

$$\begin{aligned} \langle \log a_{i,j}^{(c,k)} \rangle &= \sum_{h=1}^{j-1} \left[\psi(\tilde{\alpha}_{i,h}^{(c,k)}|_2) - \psi(\tilde{\alpha}_{i,h}^{(c,k)}|_1 + \tilde{\alpha}_{i,h}^{(c,k)}|_2) \right] \\ &\quad + \left[\psi(\tilde{\alpha}_{i,j}^{(c,k)}|_1) - \psi(\tilde{\alpha}_{i,j}^{(c,k)}|_1 + \tilde{\alpha}_{i,j}^{(c,k)}|_2) \right] \\ i = 0, \dots, I; j = 1, \dots, I. \end{aligned} \quad (13)$$

In (12), $\langle \log f(x_t^{(c,m,n)} | \theta_i) \rangle$ is the data likelihood, with expectation performed with respect to the distribution of component θ_i . For our application, the detailed derivation can be referred to [14].

- 5) Assume $q(\rho_m^{(c)}) = \text{Beta}(\tilde{f}_m^{(c)}|_1, \tilde{f}_m^{(c)}|_2)$, then

$$\begin{aligned} \tilde{f}_m^{(c)}|_1 &= 1 + \sum_{h=m}^{M_c} \langle z_{h,m}^{(c)} \rangle \\ \tilde{f}_m^{(c)}|_2 &= f_0 + \sum_{h=m}^{M_c} \sum_{e=1}^{m-1} \langle z_{h,e}^{(c)} \rangle; m = 2, \dots, M_c \end{aligned} \quad (14)$$

- 6) $q(z_m^{(c)})$ has the similar form to $q(\mathbf{s}^{(c,m,n)})$,

$$\begin{aligned} q(z_m^{(c)}) &\propto \exp \left[\langle \log u_{m,z_m^{(c)}}^{(c)} \rangle \right. \\ &\quad + \sum_{n=1}^N \left[\langle \log a_{0,s_1}^{(c,z_m^{(c)})} \rangle \right. \\ &\quad + \sum_{l=1}^{L-1} \langle \log a_{s_l^{(c,m,n)}, s_{l+1}^{(c,m,n)}}^{(c,z_m^{(c)})} \rangle \\ &\quad \left. \left. + \sum_{l=1}^L \langle \log f(x_l^{(c,m,n)} | \theta_{s_l^{(c,m,n)}}) \rangle \right] \right] \\ m = 1, \dots, M_c; c = 1, \dots, C \end{aligned} \quad (15)$$

with

$$\begin{aligned} \langle \log u_{m,k}^{(c)} \rangle &= \sum_{h=k+1}^m \left[\psi(\tilde{f}_h^{(c)}|_2) - \psi(\tilde{f}_h^{(c)}|_1 + \tilde{f}_h^{(c)}|_2) \right] \\ &\quad + \left[\psi(\tilde{f}_k^{(c)}|_1) - \psi(\tilde{f}_k^{(c)}|_1 + \tilde{f}_k^{(c)}|_2) \right]; k = 1, \dots, m. \end{aligned} \quad (16)$$

In the following experiments, we terminate the algorithm when the change in $\mathcal{L}(q(\Psi))$ is negligibly small (e.g., the lower bound variation rate between two iterations lower than 10^{-5}).

C. Classification Phase

After learning the model, classification can be carried out. A given HRRP sample under test, $\mathbf{x}^* = [x_1^*, \dots, x_L^*]^T$, is submitted to all learned HMMs (each HMM corresponds to a subframe) and the subframe-conditional likelihood can be calculated according to (1) as

$$\begin{aligned} p(\mathbf{x}^* | m, c) &= \sum_{k=1}^{M_c} \langle z_{m,k}^{(c)} \rangle \\ &\quad \cdot \left[\sum_{\mathbf{s}^*} \langle a_{0,s_1}^{(c,k)} \rangle \prod_{l=1}^{L-1} \langle a_{s_l^*, s_{l+1}^*}^{(c,k)} \rangle \prod_{l=1}^L p(x_l^* | \theta_{s_l^*}) \right] \end{aligned} \quad (17)$$

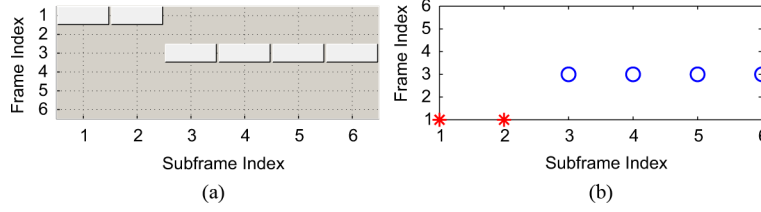


Fig. 6. Frame segmentation results via the dynamic MTL-TSB-HMM for the synthetic data. (a) Posterior expectation of the indicator variables $z_m^{(1)}$ with outcome k for $m = 1, \dots, 6$ and $k = 1, \dots, 6$. The plot is meant to visualize the probability matrix, where the white box reflects the relative probability strength (where there is no white square, the probability is zero; where there is only one white box for a subframe, the probability is one). (b) Frame labeling results of each subframe. The symbols denote ground truth: * represents the segmentation associated with subframes 1 and 2, o represents the segmentation associated with subframes 3–6; and each subframe is assigned to a frame index ($M_1 = 6$), with the index shown along the vertical axis.

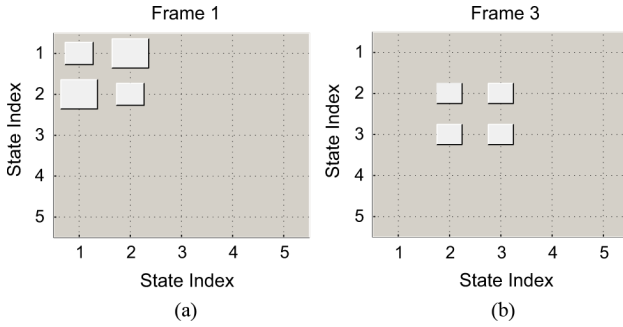


Fig. 7. Posterior expectation of state-transition matrices of each segmented frame, inferred via the dynamic MTL-TSB-HMM for the synthetic data, where the white box reflects the relative probability strength and no white square means the probability is zero. In the model a state truncation level of $I = 30$ was employed. Nevertheless, since three meaningful states (state 1, state 2, and state 3) are correctly inferred, here we only show the state-transition probabilities across the first 5 states and those of other states, which are zeros, are omitted for clarity.

where $\langle \cdot \rangle$ means the posterior expectation for the latent variable over the corresponding distribution on it, e.g., $\langle z_{m,k}^{(c)} \rangle$ denotes the posterior expectation of the probability of the m th subframe from the c th target selecting the inferred frame k and $\langle a_{s_l^*, s_{l+1}^*}^{(c,k)} \rangle$ denotes the posterior expectation of state transition probability from state s_l^* to state s_{l+1}^* for the inferred frame k and $\langle \theta_{s_l^*} \rangle$ denotes the posterior expectation of the observation model parameters associated with state s_l^* , with the corresponding state indicator for the l th range cell $s_l^* \in \{1, \dots, I\}$. All these posterior expectations can be easily calculated after learning $q(\Psi)$ as discussed in Section IV-B.

Then the class-conditional likelihood of target c ($c \in \{1, \dots, C\}$) is $p(\mathbf{x}^*|c) = \max_m p(\mathbf{x}^*|m, c)$ with $m \in \{1, \dots, M_c\}$. According to Bayes theorem, the posterior probability of target c is $p(c|\mathbf{x}^*) = \frac{p(\mathbf{x}^*|c)p(c)}{\sum_c p(\mathbf{x}^*|c)p(c)}$, where the prior class probability $p(c)$ is usually assumed to be the same for all targets of interest. The class membership of the test sample can be assigned to the class with the maximum posterior probability, i.e., $\hat{c} = \arg \max_c p(c|\mathbf{x}^*)$. Similar frame matching method is also utilized in [7], [9]. The recognition rate in the following experimental results is obtained by dividing the total number of test samples into the number of correctly classified test samples. In this way, we just utilize a single HRRP sample to determine its class membership in the classification phase. If using tracking radar, we can directly use the fusion methods discussed in [26] to make decision based on multiple HRRP samples from a target.

V. EXPERIMENTAL RESULTS

To apply the dynamic MTL-TSB-HMM model to the following synthetic data and measured HRRP data, we first complete the prior specification by choosing hyperparameter values. In particular, the prior for each $\rho_k^{(c)}$ for $k = 2, \dots, M_c$ and $c = 1, \dots, C$ is chosen to encourage the aspect-dependent subframes to be shared; consequently, we set the prior Beta(1, 5), same as the corresponding prior used in [23]. Since the parameter α_i controls the prior distribution on the number of states, we place the prior Ga($10^{-6}, 10^{-6}$) on each α_i for $i = 0, \dots, I$ to promote sparseness on states. Here we set the truncation level for stick-breaking prior at $I = 30$. Meanwhile, we use the univariate Normal distribution to describe the observation model $f(\cdot)$, i.e., $f(\theta_i) = \text{Norm}(\mu_i, \sigma_i^{-1})$ for $i = 1, \dots, I$, where Norm($\cdot$) represents the Normal distribution, μ_i and σ_i respectively denote the mean parameter and the precision parameter (inverse of variance) in the Normal distribution (here $\theta_i = \{\mu_i, \sigma_i\}$) and the corresponding prior on μ_i and σ_i is set $H(\mu_i, \sigma_i) = \text{Norm}\left(r, \frac{1}{\tau\sigma_i}\right) - \text{Ga}(b, \lambda)$ to preserve the conjugacy requirements, where r, τ, b and λ are parameters associated with the Normal and Gamma distributions in H [2]. In the following two subsections, Section V-A for synthetic data only concerns the training phase, while Section V-B for measured HRRP data concerns both training and classification phases.

A. Results on Synthetic Data

We synthesized data from the following HMM:

$$\begin{aligned}
 x^{(1,m,n)}(l) &\sim \text{Norm}(\mu_{s_l^{(1,m,n)}}, \sigma_{s_l^{(1,m,n)}}^{-1}); l = 1, \dots, 200, \\
 n &= 1, 2; m = 1, \dots, 6 \\
 \mu_1 &= 0, \mu_2 = 10, \mu_3 = -10 \\
 \sigma_1^{-1} &= 0.01, \sigma_2^{-1} = \sigma_3^{-1} = 1 \\
 \mathbf{W}^{(1,m_1)} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.25 & 0.75 \\ 0 & 0.75 & 0.25 \end{bmatrix} \\
 \mathbf{w}_0^{(1,m_1)} &= [0 \quad 0.5 \quad 0.5]; m_1 = 1, 2 \\
 \mathbf{W}^{(1,m_2)} &= \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0.5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
 \mathbf{w}_0^{(1,m_2)} &= [0.5 \quad 0.5 \quad 0]; m_2 = 3, \dots, 6.
 \end{aligned} \tag{18}$$

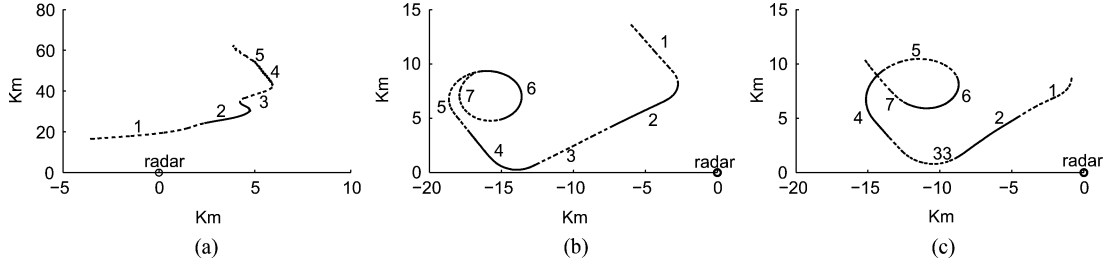


Fig. 8. Projections of target trajectories onto the ground plane. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

Since the first row of $\mathbf{W}^{(1,m_1)}$ and the third row of $\mathbf{W}^{(1,m_2)}$ contain all zero entries (not pmf as discussed in Section II), the corresponding two states do not occur in the subframes. In this example, the first 2 subframes share a kind of state-transition statistics between states 2 and 3, while the last 4 subframes share another form between states 1 and 2. In this example, the VB algorithm converges after about 40 iterations.

Fig. 6(a) presents the posterior expectation of the indicator variables $z_m^{(1)}$ with outcome k for $m = 1, \dots, 6$ and $k = 1, \dots, 6$, i.e., $\{\langle z_{m,k}^{(1)} \rangle\}_{k=1}^6$, where the white box reflects the relative probability strength and no white square means the probability is zero. Since there is only one white box for each subframe in Fig. 6(a), it appears that there is a probability of one where the white boxes are. Obviously, only 2 meaningful frames are inferred, although the initial number of frames is 6. Based on $\{\langle z_{m,k}^{(1)} \rangle\}_{k=1}^6$, we can further approximate the m th subframe's membership by assigning it to the frame with the largest probability. Fig. 6(b) presents the “hard” (most likely) decisions employed to provide the frame label (the Bayesian analysis can also yield a “soft” decision in terms of a full posterior distribution) for each subframe. To indicate the ground truth, different symbols are used to represent different segmentations in the six subframes. From Fig. 6(b), the frame segmentation result is in agreement with ground truth, i.e., the subframes 1 and 2 are associated with a frame (frame 1) and the subframes 3–6 with another frame (frame 3).

The proposed model learns the posterior state-transition matrices of each inferred frame and in so doing infers an estimate of the proper number of states. If state i is used for the inferred frame k , then the posterior probability on this state usage, $\sum_{j=1}^I \langle a_{j,i}^{(1,k)} \rangle$ should be nonzero. As shown in Fig. 7, although we initialized the truncation level to $I = 30$, our model correctly infers 3 states (state 1, state 2, and state 3) for the two segmented frames, where the posterior expectations of μ_i for $i = 1, 2, 3$ are $\langle \mu_1 \rangle = -9.9800$, $\langle \mu_2 \rangle = 9.9966$ and $\langle \mu_3 \rangle = 0.0003$; and the posterior expectations of σ_i^{-1} for $i = 1, 2, 3$ are $\langle \sigma_1^{-1} \rangle = 1.0199$, $\langle \sigma_2^{-1} \rangle = 0.9556$ and $\langle \sigma_3^{-1} \rangle = 0.0102$.

B. Results on HRRP Data

1) *Measured Data:* We examine the performance of the proposed model on the measured HRRP data from three real airplanes, which are also used in [7]–[9]. The parameters of the airplane targets and measurement radar are shown in Table I and the projections of target trajectories onto the ground plane are shown in Fig. 8, from which the aspect angle of an airplane can be estimated according to its relative position to radar. As shown

TABLE I
PARAMETERS OF PLANES AND RADAR IN THE ISAR EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

in Fig. 8, the measured data are segmented, where each segment contains 25 600 HRRP samples and each sample is a 256-length vector, i.e., $L = 256$. With large SNR, the effect of noise in the measured data on the recognition can be ignored.

In order to test the generalization performance of the recognition methods, there are two requirements for choosing the training data and test data. a) The training data cover almost all of the target-aspect angles of the test data; b) the elevation angles of the test data are different from those of the training data. Therefore, data from the second segment and a part of the fifth segment of Yark-42, the sixth and the seventh segments of Cessna Citation S/II, the fifth and the sixth segments of An-26 are taken as the training samples and other data segments are taken as test samples in our experiments. There are totally 35 840 HRRP samples from Yark-42, 51 200 from Cessna Citation S/II, and 51 200 from An-26 used as training data.

2) *Data Preprocessing:* As discussed in [8] and [9], it is a prerequisite for radar HRRP target recognition to deal with the target-aspect, time-shift, and amplitude-scale sensitivity problems.

We have discussed the target-aspect sensitivity issue in Section III-A. It is important to state that the initial subframes for our model should be much smaller than the aspect-frames used in [7]–[9], which are similarly obtained via the equal interval partition approach as shown in Fig. 2. In our previous work [7]–[9], there are 35 aspect-frames for Yark-42, 50 for Cessna Citation S/II and 50 for An-26, of which each aspect-frame has 1024 HRRP samples; while in the following experiments, each subframe has 256 HRRP samples, i.e., $N = 256$ and there are totally 540 subframes, i.e., $\sum_{c=1}^3 M_c = 540$.

Since an HRRP sample is only a part of received radar echo extracted by a range window, in which target signal is included, the position of the target signal in an HRRP sample can vary

TABLE II

CONFUSION MATRICES AND AVERAGE RECOGNITION RATES OF THE INDEPENDENT MTL-TSB-HMM AND THE DYNAMIC MTL-TSB-HMM, FOR 32×540 TRAINING DATA. THE FORMER ONE IS THE LIMITING CASE OF OUR PROPOSED MODEL WITH $\rho_m^{(c)} \rightarrow 1$ FOR $m = 1, \dots, M_c$ AND $c = 1, 2, 3$, I.E., THE STATE-TRANSITION STATISTICS ARE INDEPENDENT

	independent MTL-TSB-HMM			dynamic MTL-TSB-HMM		
	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26
Yark-42	86.67	11.58	1.75	92.58	6.34	1.08
Cessna Citation S/II	4.75	82.30	12.95	2.80	90.60	6.60
An-26	3.35	3.50	93.15	3.45	1.40	95.15
average recognition rates	87.37			93.67		

with the measurement. This is the time-shift sensitivity. Nevertheless, since the HMM structure can describe the sequential dependence, the time-shift compensation is not needed for the classification phase. Thus the slide correlation processing [25] is used here just for each subframe in the training phase.

Furthermore, the intensity of an HRRP is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses and so on, therefore, HRRPs measured by different radars or under different conditions will have different amplitude-scales. To deal with this amplitude-scale sensitivity, each HRRP sample is normalized by dividing its L_2 norm.

In the following experiments via our dynamic MTL-TSB-HMM, in the training phase, the HRRPs in each subframe are time-shift compensated and amplitude L_2 normalized; while in the classification phase, the single normalized HRRP sample is tested one by one and the whole test dataset is the same as those used in [7] and [9].

3) *Recognition Performance*: The recognition accuracies of the dynamic MTL-TSB-HMM and the independent MTL-TSB-HMM versus the size of training dataset are shown in Fig. 9, of which the latter one is the limiting case of our proposed model with $\rho_m^{(c)} \rightarrow 1$ for $m = 1, \dots, M_c$ and $c = 1, 2, 3$, i.e., the state-transition statistics are independent. Here we utilized the six training datasets: 1×540 , 2×540 , 4×540 , 8×540 , 16×540 , and 32×540 , where 8×540 means that 8 HRRP samples randomly selected from each of the 540 subframes, i.e., there are totally 4320 training samples, similarly for other data size expressions. The subframe segmentation approach discussed in Section V-B-II is adopted for the dynamic MTL model, while the previous aspect-frame segmentation approach in [7]–[9] for the independent MTL model, i.e., the four initial subframes for the former model corresponds to one aspect-frame for the latter model (totally 135 aspect-frames are used for the independent MTL model and 540 subframes for the dynamic MTL model). It is worth pointing out that we only use five runs' results for each training dataset here to evaluate the recognition performance due to the intensive computation for training, which is further analyzed in Section V-B-VI. The recognition rates in Fig. 9 are averaged over these five runs and the error bars denote the corresponding standard deviations. Table II compares the confusion matrices and average recognition rates of the two models for 32×540 training data, respectively, generated via a run. From Fig. 9 and Table II, we can clearly see that the dynamic MTL model

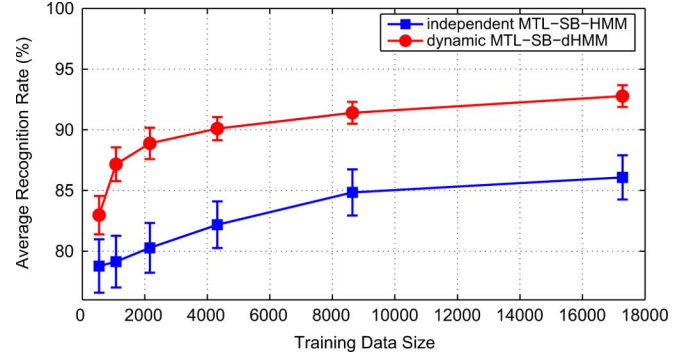


Fig. 9. Recognition accuracies of the dynamic MTL-TSB-HMM and the independent MTL-TSB-HMM versus the size of training dataset. The latter one is the limiting case of our proposed model with $\rho_m^{(c)} \rightarrow 1$ for $m = 1, \dots, M_c$ and $c = 1, 2, 3$, i.e., the state-transition statistics are independent. The recognition rates are averaged over five runs and the error bars denote the corresponding standard deviations.

has better recognition performance than the independent MTL model and the average recognition rate of the proposed model is about 5 percentage points larger than that of its limiting case with the maximum innovation for a given training dataset. In addition, the larger training dataset generally yields the higher and stabler recognition rate. Nevertheless, at the same time, the larger training dataset also needs the huger computation burden. Therefore, in a real application, we need to consider the system resource and strike an appropriate balance between recognition performance and computation burden. The detailed computation burden comparison is given in Section V-B-VI.

According to the introduction in Section V-B-I, the SNR in the measured data is large. For aircraft-like targets under a look-up scenario, the noises in the inphase and quadrature echoes of targets can be assumed to be Gaussian white noises. In order to evaluate the effect of noises on the proposed model and other existing statistical models, we add simulated white noises to the inphase and quadrature components of the raw test data. The SNR is defined as

$$\begin{aligned}
 \text{SNR} &= 10 \times \log_{10} \left(\frac{\bar{P}_{\mathbf{x}}}{P_{\text{Noise}}} \right) \\
 &= 10 \times \log_{10} \left(\frac{\sum_{l=1}^L P_{x_l}}{L \times P_{\text{Noise}}} \right) \quad (19)
 \end{aligned}$$

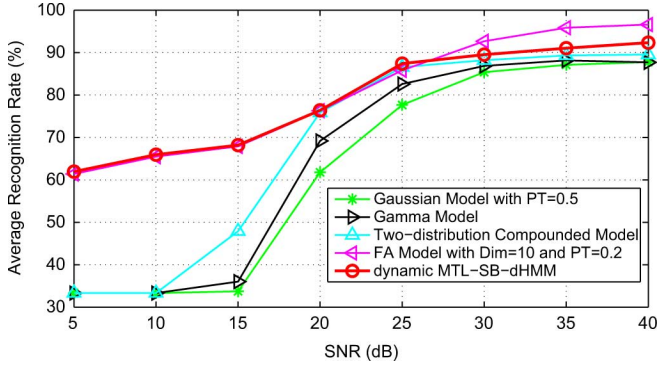


Fig. 10. Variation of the recognition performance with SNR, via Gaussian model with power transformation ($\alpha = 0.5$) [13], [18], Gamma model [6], our two-distribution compounded statistical model [9], FA model with the dimensionality of the latent factor variable $D = 10$ and power transformation ($\alpha = 0.2$) [7] and the proposed dynamic MTL-TSB-HMM. Here Gaussian model, Gamma model, the two-distribution compounded model and our dynamic MTL model are trained with $32 \times 540 = 17280$ training data; while the FA model with $1024 \times 135 = 138240$ training data. The recognition rates are averaged over five runs.

where $\bar{P}_x$ denotes the average power of original HRRP, P_{x_l} denotes the power of the original echo per range cell, L denotes the number of range cells (here $L = 256$), and P_{Noise} denotes the power of noise. Thus $10^{\frac{\text{SNR}}{10}}$ equals to the rate of the average power of original HRRP to the power of the noise. Fig. 10 depicts the average recognition rates versus SNR, generated via the five statistical models, i.e., Gaussian model [13], [18], Gamma model [6], the two-distribution compounded model [9], FA model [7] and the new dynamic MTL model. Obviously, the Gaussian model with power transformation is most sensitive to noise. When $\text{SNR} \geq 30$ dB, the FA model with power transformation outperforms other four models; when SNR is 20 or 25 dB, our two-distribution model, FA model, and dynamic MTL model obtain the similar performance; when $\text{SNR} \leq 15$ dB, only the FA model and the dynamic MTL model can yield recognition rates larger than 60%. Although the comparable recognition accuracies got by the two models (especially with the low SNR), it is important to point out that the FA model requires much more training data than our new model, due to the full covariance estimation for the 256-length HRRP vector. In the recognition experiments, Gaussian model, Gamma model, the two-distribution compounded model, and our dynamic MTL model are trained with $32 \times 540 = 17280$ training data; while the FA model with $1024 \times 135 = 138240$ training data.

4) *Results Analysis:* As discussed in Section V-A, the proposed nonparametric model can infer the proper numbers of HMM states and segmented frames for the training data. Table III summarizes the corresponding model selection results for the different training data. Generally, the number of meaningful states is less than 25 and that of segmented frames less than 45, where the meaningful state or segmented frame means that the posterior probability on the state or frame usage is nonzero.

In Section V-A, we show the posterior expectation of transition matrix of each inferred frame, i.e., $\langle A^{(1,k)} \rangle$ with $k = 1, 2$,

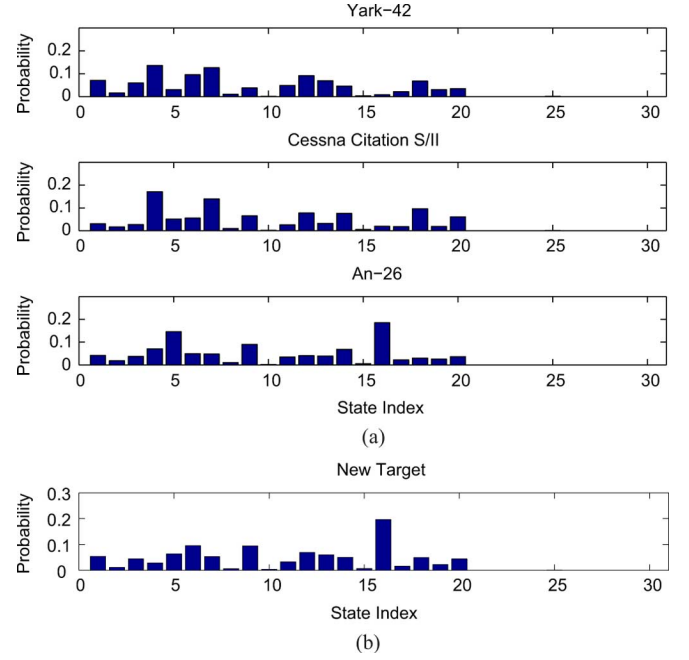


Fig. 11. Hidden state distribution across all plane targets for 32×540 training data, generated via a run of the dynamic MTL-TSB-HMM. (a) Three learned targets, Yark-42, Cessna Citation S/II, and An-26. (b) New target, a civil airplane, of which the HMM model is predicted based on the learned dynamic MTL model associated with (a). In the example, a state truncation level of $I = 30$ was employed. Note all these 30 hidden states are shared across the 17 280 training HRRP data from the three targets and 21 meaningful states are inferred.

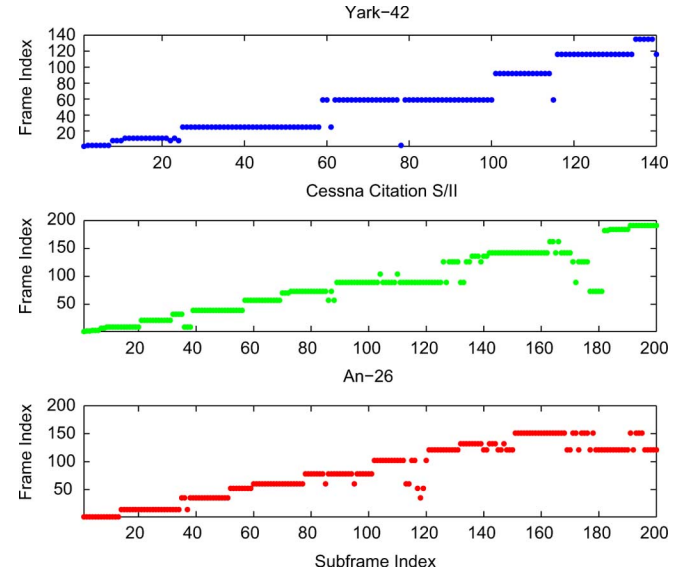


Fig. 12. Frame labels of all 540 subframes for 32 training samples per sub-frame, generated via a run of the dynamic MTL-TSB-HMM. Here each sub-frame is assigned to a frame index ($M_1 = 140$ for Yark-42, $M_2 = 200$ for Cessna Citation S/II, and $M_3 = 200$ for An-26), with the index shown along the vertical axis. There are nine meaningful frames inferred for Yark-42, 20 for Cessna Citation S/II, and 10 for An-26.

for the synthetic example in Fig. 7. Here for brevity (much more frames inferred for the real data), Fig. 11(a) presents the posterior state distribution for each of the three learned targets, i.e., $\sum_{m=1}^{M_c} \sum_{k=1}^m \langle z_{m,k}^{(c)} \rangle \left(\sum_{j=1}^I \langle a_{j,i}^{(c,k)} \rangle \right)$ for $i = 1, \dots, I$ and $c = 1, 2, 3$, generated by a run of our dynamic model with

TABLE III
NUMBERS OF MEANINGFUL STATES AND SEGMENTED FRAMES FOR DIFFERENT TRAINING HRRP DATA, INFERRED VIA THE DYNAMIC MTL-TSB-HMM. HERE WE GIVE THE AVERAGE RESULTS OVER FIVE RUNS FOR EACH TRAINING DATA SIZE

training data size	1×540	2×540	4×540	8×540	16×540	32×540
average number of states	18.4	21.6	22.4	22.8	21.6	23.2
average number of frames	23.6	23.6	29.2	30.6	35.4	40.6

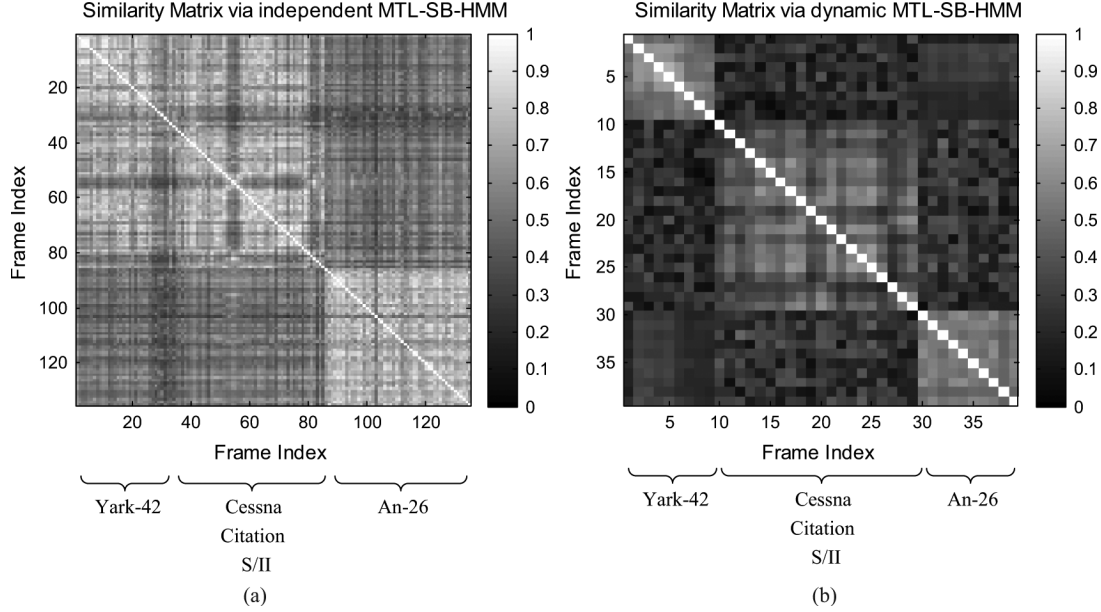


Fig. 13. Similarity matrices based on the posterior state-transition matrices across equally segmented frames or inferred frames for 32×540 training data, generated respectively via a run of the independent MTL-TSB-HMM and the dynamic MTL-TSB-HMM. The former one is the limiting case of our proposed model with $\rho_m^{(c)} \rightarrow 1$ for $m = 1, \dots, M_c$ and $c = 1, 2, 3$, i.e., the state-transition statistics are independent: (a) Independent MTL-TSB-HMM. (b) Dynamic MTL-TSB-HMM.

32×540 training data. Although we initialized the truncation level to $I = 30$, there are only 21 states with the posterior state usage larger than zero and those of other 9 states are zero, which justifies using the truncated version stick-breaking prior for our data. In addition, the underlying 21 states are shared over all three targets with the different probabilities of usage, which shows the utility of using MTL.

Fig. 12 shows the segmented frame labels for the three targets, also generated by a run of our dynamic model with 32×540 training data. Here we still use the “hard” decisions discussed in Section V-A. Obviously, the lengths of the inferred segmentations are different and the distant repetition or “innovation” occurs at some aspects, which accords with our model assumption. There are nine meaningful frames inferred for Yark-42, 20 for Cessna Citation S/II, and 10 for An-26. Please note the number of total subframes is 540, i.e., $\sum_{c=1}^3 M_c = 540$ and the inferred number of frames is much less than 135, which is the number of frames used for other models[7]–[9], therefore, the computation burden in the classification phase can be reduced.

Within the MTL framework, the set of HMM states are shared (with different posterior state-transition statistics) across subframes and similar subframes should share similar posterior state-transition distributions over these states. To measure the

similarity of the inferred frames k_{c_1} and k_{c_2} , we define the similarity measure

$$\text{Sim}(k_{c_1}, k_{c_2}) = \frac{\sum_{i=1}^I \sum_{j=1}^I (a_{i,j}^{(c_1, k_{c_1})} - \bar{a}_{c_1}) \cdot (a_{i,j}^{(c_2, k_{c_2})} - \bar{a}_{c_2})}{\| \mathbf{A}^{(c_1, k_{c_1})} - \bar{a}_{c_1} \cdot \mathbf{1}_{I \times I} \|_2 \| \mathbf{A}^{(c_2, k_{c_2})} - \bar{a}_{c_2} \cdot \mathbf{1}_{I \times I} \|_2};$$

$$c_1, c_2 = 1, 2, 3; k_{c_1} \in \{1, \dots, M_{c_1}\}; k_{c_2} \in \{1, \dots, M_{c_2}\} \quad (20)$$

where $\bar{a}_{c_o} = \frac{1}{I^2} \sum_{i=1}^I \sum_{j=1}^I a_{i,j}^{(c_o, k_{c_o})}$ with $o = 1, 2$, $\mathbf{1}_{I \times I}$ denotes an $I \times I$ matrix of ones and $\| \mathbf{B} \|_2$ denotes the L_2 norm of matrix $\mathbf{B}$. Fig. 13(a) and (b) presents the similarity matrices for the examples associated with Table II, where we only measure the similarities between the segmented frames inferred via the dynamic model in Fig. 13(b). Compared to the result generated via the independent MTL model, the dynamic MTL model greatly decreases the similarities between different frames, therefore, the segmentation via our dynamic model is more reasonable.

5) *Model Prediction for New Target:* We also note that this MTL model, by sharing states across the training targets, will

TABLE IV

CONFUSION MATRIX AND AVERAGE RECOGNITION RATE OF THE DYNAMIC MTL-TSB-HMM, FOR 32×540 TRAINING DATA FROM YARK-42, CESSNA CITATION S/II, AN-26 AND 1500 TRAINING DATA FROM THE NEW TARGET (A CIVIL AIRPLANE). THE HMM MODEL OF THE NEW TARGET IS PREDICTED BASED ON THE LEARNED DYNAMIC MODEL ASSOCIATED WITH TABLE II

	Yark-42	Cessna Citation S/II	An-26	New Target (Civil Airplane)
Yark-42	92.58	6.34	1.08	0
Cessna Citation S/II	2.55	90.35	6.60	0.50
An-26	3.25	1.35	94.65	0.75
New Target (Civil Airplane)	0.40	6.80	27.40	65.40
average recognition rates	85.75			

not only model any one of the training targets, but also infer relationships between the multiple training targets within this category, i.e., aircraft here. This generalization is beneficial when testing on targets not seen while training, as long as the test targets of interest are within the category of training data. Here since the HMM states are shared, if we can predict the state-transition matrix for a new airplane target, then the corresponding HMM model is obtained.

In the VB inference engine, for a new training subframe $\{x_l^{(*,1,n)}\}_{l=1, n=1}^{L, N^*}$, we can predict its posterior state-transition matrix as

$$\begin{cases} a_{0,j}^* \propto \sum_{n=1}^{N^*} \langle s_{1,j}^{(*,n)} \rangle \\ a_{i,j}^* \propto \sum_{n=1}^{N^*} \sum_{l=1}^{L-1} \langle s_{l,i}^{(*,n)} \rangle \langle s_{l+1,j}^{(*,n)} \rangle \end{cases}; i, j \in \{1, \dots, I\}$$

$$q(\mathbf{s}^{(*,n)}) \propto \prod_{l=1}^L \log f(x_l^{(*,1,n)} | \langle \theta_{s_l^{(*,n)}} \rangle). \quad (21)$$

In application, we only consider the meaningful states inferred via the learned model.

To evaluate the prediction performance, we use a measured HRRP sequence from a civil airplane as the new target data of interest. The radar parameters are the same as those in Table I. The length of the new data sequence is 2000, of which we select one HRRP sample from every four samples as a test data and the rest as the training data, thus there are 1500 training data and 500 test data, i.e., $N^* = 1500$. Here the training data are used to predict the HMM parameters of the new target based on the learned MTL model from the three training targets via (21) and the test data are used in the classification phase. Fig. 11(b) depicts the posterior state distribution of the new model, predicted based on the learned dynamic model associated with Table II. The usage of the hidden state is different from those for the learned three targets in Fig. 11(a). The confusion matrix and average recognition rate, generated via the learned dynamic model and the corresponding predicted HMM for the new target, are presented in Table IV. By introducing the new target, the recognition performance of the learned three targets doesn't change much and the recognition accuracy of the predicted target is larger than 65%. Although the recognition accuracy, 65.40%, is not as high as those of the trained targets, the complicated training phase is omitted for the new target. Such method can be used for the on-line learning.

6) *Computation Burden*: All experiments have been performed in nonoptimized software written in Matlab, on a Pentium PC with 1.50 GHz CPU and 2G RAM. For the measured HRRP data, in the training phase, the independent MTL-TSB-HMM and the dynamic MTL-TSB-HMM, both with truncation level $I = 30$, respectively required about 15 and 30 h (processing the 35 840 training samples from all targets simultaneously). The computation time given here is averaged over 30 runs with different training data sizes.

Although the above computation is pretty expensive, we know that the computation cost in the training phase can be ignored for an off-line learning (or training) system and it is more important to evaluate the computation cost in the classification phase. Compared to the independent MTL (training data are equally segmented into 135 aspect-frames), much fewer frames are inferred via the dynamic MTL, as shown in Table III and Fig. 12, therefore, the time cost to match a test sample with all frame models, i.e., the computation cost in the classification phase, via the latter algorithm should be much fewer than that via the former one. Averagely, by our experiments, in the classification phase, the independent MTL-TSB-HMM and the dynamic MTL-TSB-HMM required 1.3810 and 0.4709 s to match a test sample with all frame models, respectively, which means that the computation cost of the independent MTL model is nearly three times larger than that of the dynamic MTL model in the classification phase. Therefore, our algorithm can effectively improve the on-line recognition speed. In addition, the prediction of the new target's model only required 13.1925 s for the example shown in Table IV, i.e., for the 1500 new training samples from the new target.

VI. CONCLUSION

We have developed a dynamic Bayesian spatiotemporal multitask learning model for radar HRRP data, termed the dynamic multitask learning based HMM with stick-breaking prior (dynamic MTL-TSB-HMM). The new model is motivated by [23], which are proposed for music analysis and the proposed construction allows the HMM states shared for all tasks (here the multiple tasks are linked to specific subframes of HRRPs) and the convenient VB inference, of interest for the large-scale motivating RATR problem. For radar HRRP target recognition, the spatial structure across range cells is described by the hidden Markov structure and the temporal dependence between HRRP samples is described by the time evolution of the transition probabilities. The algorithm has been demonstrated on synthetic data

and measured HRRP data. The RATR experiments show that the proposed model not only has better recognition performance than the independent MTL-TSB-HMM model (the limiting case of the proposed model with the maximum innovation) but also is more robust to noise than other existing statistical models for radar HRRP data except for the FA model. Our new model achieves comparable recognition accuracy to the FA model (especially with the low SNR), yet requires substantially fewer training data. Furthermore, since all HMM states are shared across targets, we can simply expect the corresponding HMM model for the new targets not seen while training via the proposed MTL model.

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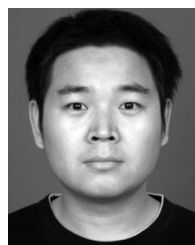
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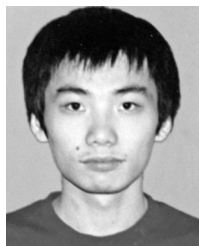
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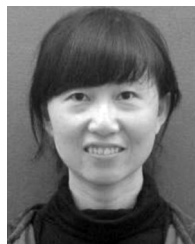
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